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SHORT-PAPER

Content Moderation on Social Media in the EU: Insights From the DSA Transparency Database

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Content Moderation on Social Media in the EU: Insights From the DSA Transparency Database

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ABSTRACT

The Digital Services Act (DSA) represents a major legislative framework that mandates large social media providers to file *Statements of Reasons* (SoRs) to the DSA Transparency Database whenever they remove or restrict access to certain content on their platforms in the EU. In this work, we empirically analyze this unique data source and provide an early look at content moderation decisions of social media platforms in the EU. Our empirical analysis based on more than 156 million SoRs reveals significant differences in content moderation practices and how large social media platforms implement their obligations under the DSA. Our findings have important implications for regulators, suggesting the need to lay out more specific rules that ensure common standards on how social media providers handle rule-breaking content on their platforms.

CCS CONCEPTS

• **Human-centered computing** → **Social media**; **Empirical studies in collaborative and social computing**; • **Applied computing** → **Sociology**.

KEYWORDS

Content moderation, social media, online harms, DSA, EU

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1 INTRODUCTION

Social media platforms have become essential channels for accessing information that offer various benefits but also facilitate the dissemination of harmful or illegal content (e. g., hate speech, calls for violence, disinformation). Concerns about such content's impact on elections public health, and safety have grown recently [e. g., 1, 3]. In response, platforms employ diverse content moderation systems, i. e., mechanisms that aim to prevent harm by removing or reducing the visibility of rule-breaking content [5]. However,

their degree of strictness varies greatly between platforms, with each platform keeping the specifics of how it enacts its moderation decisions largely opaque [7]. Accordingly, there have been increasing calls for legislation to revise the present model of self-regulation, where social media platforms primarily define the rules and procedures of online content moderation [10, 11].

The EU, recognizing the need for greater control and oversight, has recently introduced the Digital Services Act (DSA) that establishes a new set obligations for social media providers [10, 11]. Major platforms can now be held responsible for the risks illegal content on their platforms poses to society. The DSA aims to create a harmonized legal framework that avoids inconsistencies and uncoordinated procedures in content moderation. For this purpose, providers of large social media platforms are required to file *Statements of Reasons* (SoRs) explaining why content was moderated, by reference to the specific legal provision infringed. To ensure scrutiny of content moderation decisions and transparency for both platforms and users, all SoRs are made publicly available by the EU via the DSA Transparency Database (DSA-TDB).

Research goal: Here, we provide a holistic early look at the DSA-TDB. Due to this unique data source, we are, for the first time, able to empirically analyze real-world content moderation decisions of social media platforms in the EU. Specifically, we are interested in what types of content are typically moderated (**RQ1**), what the specific reasons for content moderation are (**RQ2**), to what extent content moderation is automated by the platforms (**RQ3**) and what types of moderation actions the platforms implement (**RQ4**).

Data & Methods: To answer our research questions, we collected all SoRs submitted to the DSA-TDB by major social media platforms within the first two months after its launch on September 25, 2023. Based on over 156 million SoRs, each representing a single content moderation action, we extracted and analyzed a wide variety of variables (e. g., content types, legal grounds) to understand content moderation decisions in the EU. Additionally, we implemented regression analysis to characterize which types of content moderation decisions are more likely to be performed automatically (i. e., without human intervention) by social media platforms.

Contributions: Our work presents the first empirical analysis of the EU's DSA-TDB, yielding the following key findings: (i) There are vast differences in the frequency of content moderation across platforms, with TikTok performing over 350 times more moderation decisions per user than X. (ii) Text and videos are most commonly moderated, while images and other formats undergo less frequent moderation. (iii) Primary reasons for moderation include content outside platform scope, illegal/harmful speech, and pornography, while moderation for misinformation is relatively rare. (iv) Automated methods predominantly detect and decide

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upon rule-breaking content. (v) Content moderation actions vary substantially across platforms, with some favoring post removal and others visibility reduction. Overall, our study highlights inconsistencies in how platforms implement their obligations under the DSA, suggesting the need to lay out more specific rules that ensure common standards on how providers handle rule-breaking content.

2 BACKGROUND

Content moderation: Content moderation aims to prevent the spread of illegal and undesirable content in online communities [5]. Social media platforms use various measures like content removal, visibility reduction, labeling, or account actions to address rule-breaking content [8]. Effective moderation is crucial for social network functionality, promoting guideline compliance, and reducing incivil behavior [4, 6]. Over the last couple of years, content moderation systems on mainstream platforms have become increasingly sophisticated, evolving from relatively small teams overseeing the content to the usage of automated systems that detect and intervene when rule-breaking behavior occurs [4]. Notably, each social media platform has developed various systems to implement these processes [4]; yet each platform keeps the specifics of how it enacts its moderation decisions opaque [7].

Digital Services Act: The Digital Services Act (DSA) stands as a pivotal legislative framework crafted by the EU, which aims to modernize the digital landscape and ensure safer, more open platforms [10, 11]. Adopted in July 2022 and effective since November 16, 2022, it imposes requirements on online platform providers. All Very Large Online Platforms (VLOPs) with over 45 million users in the EU (corresponding to 10% of the overall population) must meet DSA obligations since August 25, 2023. This includes regular transparency reports detailing content moderation measures, user reports, error rates of automated systems, and team qualifications (Article 15). Furthermore, as mandated by Article 17 of the DSA, VLOPs are required to provide detailed Statements of Reasons (SoRs) for any content moderation activity. The intention is to inform users about content moderation decisions and explain the underlying reasons. In accordance with Article 24(5) of the DSA, all submitted statements are to be collected and made publicly available on the DSA Transparency Database (DSA-TDB). SoRs are to be clear and specific and easily comprehensible and as precise and specific as reasonably possible under the given circumstances.

3 DATA

We downloaded *all* SoRs from the website of the DSA-TDB, that were submitted between the introduction of the database on September 25, 2023, and November 25, 2023, i. e. within an observation period of two months. In total, more than 550M SoRs were transmitted during the observation period. As we focus on content moderation on social media, we only included SoRs submitted by large social media platforms, namely *Facebook*, *Instagram*, *YouTube*, *TikTok*, *Snapchat*, *X*, *LinkedIn* and *Pinterest*. Furthermore, we excluded SoRs for content published before August 25, 2023, as the obligations associated with the DSA framework became effective from this date. The resulting dataset contains more than 156M SoRs, each including information on one content moderation action.

In our data, the largest number of SoRs was submitted by TikTok (100.15M; 64.09%), followed by Facebook (33.70M; 21.56%), Pinterest (12.45M; 7.97%), YouTube (5.12M; 3.28%), and Instagram (3.95M; 2.53%). Snapchat (0.61M; 0.39%), X (0.27M; 0.17%), and LinkedIn (0.03M; 0.02%) submitted less than one million SoRs during our observation period. Evidently, TikTok moderates substantially more content than all other platforms, some of which are much more relevant in the European market in terms of user numbers (e. g., YouTube, Facebook and Instagram).

4 EMPIRICAL ANALYSIS

4.1 Content Types (RQ1)

When submitting a SoR, platforms have to indicate the type of content that was moderated (i. e., text, images, videos, etc.). Studying the platform-specific distribution of how the platforms specify the type of the moderated content, we find considerable differences (see Fig. 1). While TikTok primarily focuses on **Text** (57.14%) and **Videos** (33.97%), Snapchat moderated a larger share of **Videos** (63.92%) and **Images** (16.96%). The platform X seems to limit its content moderation activities mainly to **Synthetic Media** (99.83%). The remaining platforms each categorized more than 50% of all content moderation actions as content type **Other**, i. e., content types that are not predefined by the DSA.¹ Altogether, we find that the type of content subject to moderation is, in many cases, strongly related to the content that is predominantly published on the respective platform (e. g., Video and Image on Snapchat, Video and Advertisement on YouTube, Pins on Pinterest, job-related content on LinkedIn). It is also worth noting that the content type is specified by the platform (i. e., self-reported), and each content piece is assigned to a single type. However, social media content is often a mix of different types. In a similar vein, AI-generated content may be classified as image/text/video or as synthetic media. Overall, it seems likely that the content type reported by platforms often describes only one dimension of a social media post.

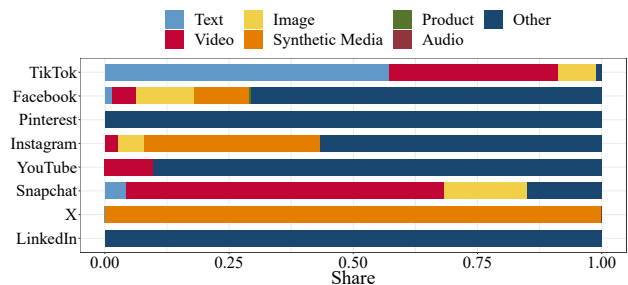


Figure 1: Distribution of moderated content types.

4.2 Reasons for Moderation (RQ2)

Platforms must specify if content is incompatible (Article 17(3)(d) DSA) or illegal (Article 17 (3)(e) DSA) when explaining moderation actions. Incompatibility dominates (99.80%), while only X focusing more than 99% on illegal content. Additionally, platforms assign

¹The majority of this content concerns violations that affect entire accounts/profiles, platform-specific content types (e. g., pins or boards on Pinterest), and (job) advertisements (in particular on Youtube, Pinterest, and LinkedIn).

one of 14 categories defining the reason for moderation. We find that there is a relatively high similarity across most platforms (see Fig. 2). With the exceptions of X and Pinterest, all platforms moderated a large proportion of content that does not correspond to the **Scope of Platform Service**, **Illegal/Harmful Speech**, and **Violence**. In contrast, Pinterest focused primarily on **Pornography/Sexualized Content** (80.95%). In the case of X, the vast majority of content moderation actions were attributed to **Violence** (41.18%) or **Pornography/Sexualized Content** (44.37%). Snapchat is the platform that moderated content in the widest variety of categories (e.g. 16.69% in **Scams & Fraud** and 10.77% **Unsafe & Illegal Products**).

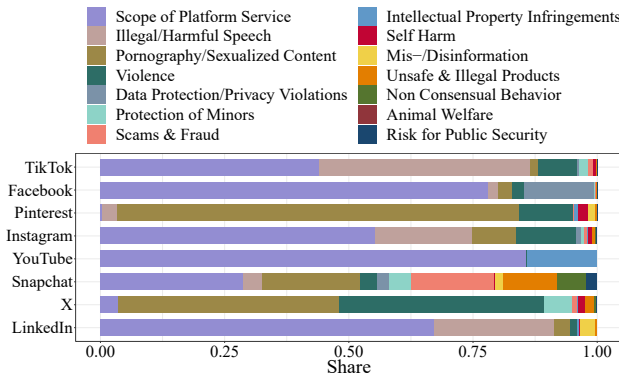


Figure 2: Distribution of reasons for moderation.

4.3 Automation of Content Moderation (RQ3)

Regarding content moderation, platforms must specify if the decision was automated (fully, partially, not), and if the violation was detected automatically (yes, no). Overall, 60.67% of all content moderation decisions were performed fully automated, 31.48% partially automated, and 7.74% not automated. Furthermore, automated detection often led to automated decisions, while manually moderated content was more frequently identified or reported by humans (e.g., users or moderators). Across all fully automated content moderation decisions, over 99% of the content was automatically identified; whereas in the partially automated and not automated categories, this proportion drops to 76.06% and 71.75%, respectively.

Across platforms, we observe further differences (see Fig. 3). While TikTok mainly employs **fully automated** decision-making, Facebook, Pinterest and Instagram tend to combine automated means and human intervention (i. e., **partially automated**). In contrast, YouTube, Snapchat, X and LinkedIn performed the majority of their content moderation **not automated**. X again stands out with a completely manual content moderation (i. e., not automated).

Regression analysis: To better understand situations in which content moderation is more likely to be performed automatically, we implement a logistic regression model estimating how the likelihood of automated content moderation decision varies across different content types and decision grounds (see Fig. 4). We find that the odds of a content moderation decision being performed automatically are $e^{3.470} \approx 32.143$ times higher if the content was detected automatically ($p < 0.01$). Across content types, we find that

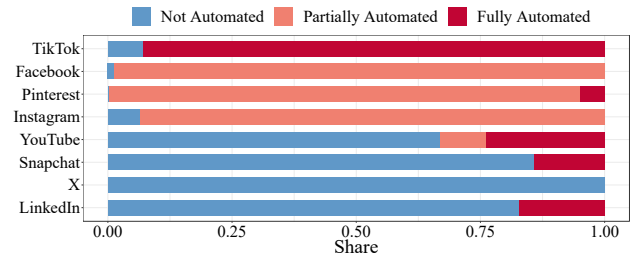


Figure 3: Distribution of automation level across platforms.

the odds of a content moderation decision performed automatically are significantly higher for text compared to other content (i. e., *Product, Synthetic Media, Audio, Other*) ($p < 0.01$). At the same time, we find a smaller positive association for *Images* ($p < 0.01$), while *Videos* are less likely to be moderated automatically ($p < 0.01$). With regard to legal grounds, we find that all categories are statistically significantly less likely to be automatically moderated compared to content that does correspond to the *Scope of Platform Service* (each $p < 0.01$). We also observe a small negative association between the time elapsed between the date the content was published on the platform and the moderation date ($p < 0.01$).



Figure 4: Coefficient estimates (circles) and 99% confidence intervals (bars) for a logistic regression model with *Automated Decision* (= 1 if yes; otherwise = 0) as DV ($n = 156,378,199$). Monthly and platform fixed effects are included. EVs are *Automated Detection* (ref. No), *Content Type* (ref.: *Other*), *Reason Moderation* (ref. *Scope of Platform Service*), and *Delay* (days elapsed until moderation).

4.4 Content Moderation Actions (RQ4)

Reported content moderation actions can be grouped into two categories: (i) actions that affect a specific content piece (i. e., **removal**, **labelling**, **disabling**, **demotion** and **age restriction** of content); (ii) actions that affect an account (i. e., **suspension** or **termination** of an account). The platforms describe in the SoRs how the content was moderated, either by selecting one of these predefined actions or by selecting **Other**. In the case of the latter, platforms can provide a short description of the action that was taken. To accommodate such cases, we employed string matching to assign the text

descriptions to the predefined action categories.² If a description is missing or unclear, the category **Other** is used (0.36%).

Overall, we observe that the most frequent types of content moderation are the **removal of content** (55.15%), the **demotion of content** (25.15%), and the **suspension/termination** (14.96%). The tendency to focus on these actions as the primary choice of content moderation is prevalent across most platforms. However, there are also differences. For instance, Pinterest puts a stronger focus on **content demotion**. Conversely, Snapchat implemented **content disabling** in over 50% of its content moderation actions. X does not distinguish between **removal** and **demotion** in its reporting to the DSA-TDB.

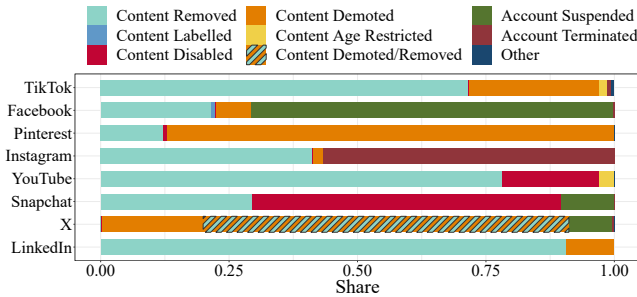


Figure 5: Distribution of *decision types* across platforms.

5 DISCUSSION & FUTURE WORK

Relevance: Effective content moderation is vital for preventing the dissemination of illegal and undesirable content in social networks [4]. However, social media providers' moderation decisions are often perceived as opaque. Previous research, primarily based on artificial environments or limited observational datasets [e. g., 2, 9], faced challenges in systematically collecting and analyzing these decisions. To foster transparency and accountability, the DSA mandates major social media platforms in the EU to make key information on their content moderation decisions publicly accessible. Utilizing the EU's DSA-TDB – a unique and previously unavailable data source – we shed light on how major social media providers moderate user-generated content on their platforms.

Implications: Our study reveals significant differences and inconsistencies in the content moderation practices of major social media platforms, with TikTok performing over 350 times more content moderation decisions per user than X. While platforms with higher moderation volumes may or may not encounter rule-breaking content more frequently, our findings indicate that different platforms focus on distinct types of rule-breaking content. Although all platforms consistently address violence and pornography, their approach to other harms like illegal speech and misinformation varies considerably. For instance, X rarely moderates illegal speech and misinformation. Additionally, the actions taken against

²The descriptions were assigned to the predefined action categories as follows: "Limited distribution", "not eligible for recommendation", "Bounce" → **Content Demoted**; "Bounce", "Ban" → **Content Disabled**; "AddTweetAnnotation" → **Content Labelled** (ordered by descending frequency). X describes a large part of the moderated content as "not suitable for work" (NSFW) (71.20%) whereby, according to the platform's own guidelines, either a reduction in visibility (i. e., **demotion**) or **removal** of the content is implemented (see striped area in Fig. 5)

rule-breaking content differ across platforms, with some removing it frequently and others opting to reduce visibility. Overall, our study suggests varied interpretations of DSA obligations, leading to a fragmented outcome that the DSA aims to avoid. These findings have implications for regulators, urging clearer guidelines to ensure consistent standards for handling rule-breaking content.

Further, our study contributes to the ongoing debate on the role of automation in content moderation. We find that the majority of rule-breaking content is already identified and processed through automated methods. While some platforms combine automated and human moderation, others (e. g., TikTok), predominantly rely on fully automated content moderation. X stands out by exclusively relying on non-automated means, albeit moderating a significantly lower content volume. Given the massive scale of content moderation in the EU, this suggests a potential need for some level of automation to meet the DSA-imposed regulatory obligations.

Limitations and future work: Our study opens avenues for future research. Firstly, our inferences are restricted to the initial two months after the introduction of the DSA-TDB in the EU, and future work can explore how patterns change over time. Secondly, the evolving nature of content moderation efforts may reach a different steady-state with growing experience and EU rule clarifications. Thirdly, understanding the impact of content removal or visibility reduction on user behavior warrants further investigation. Lastly, analyzing the source content that has been moderated would be insightful, but such data is currently unavailable. Despite these limitations, our study contributes to the understanding of social media content moderation, providing a foundation for future policy improvements. We hope our early work stimulates more research to enhance transparency in content moderation on social media.

6 ACKNOWLEDGMENTS

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